

Improving NLSAT for Nonlinear Real Arithmetic

Research Track

Zhonghan Wang

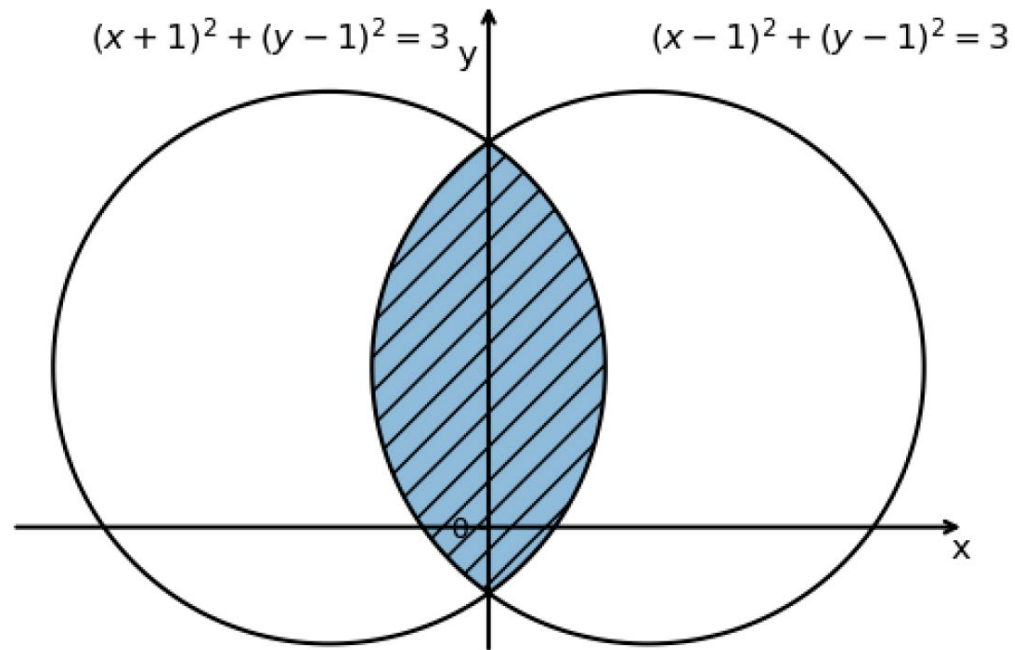
Institute of Software, Chinese Academy of Sciences

wangzhonghan272@gmail.com

November 17, 2025

Problem: SMT (QF_NRA)

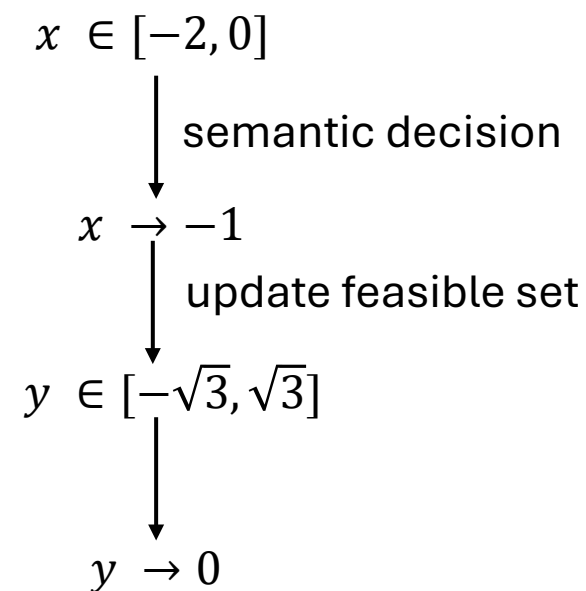
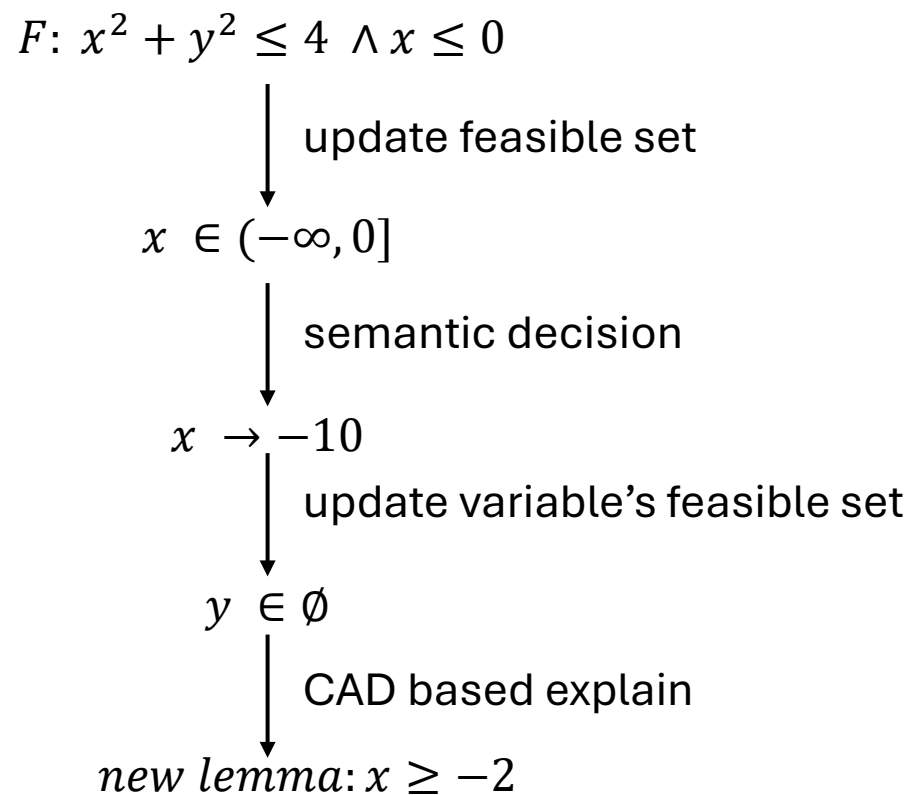
Given a Boolean combination of polynomial constraints, determine the satisfiability



Satisfied with assignment
 $\{x \rightarrow 0, y \rightarrow 1\}$

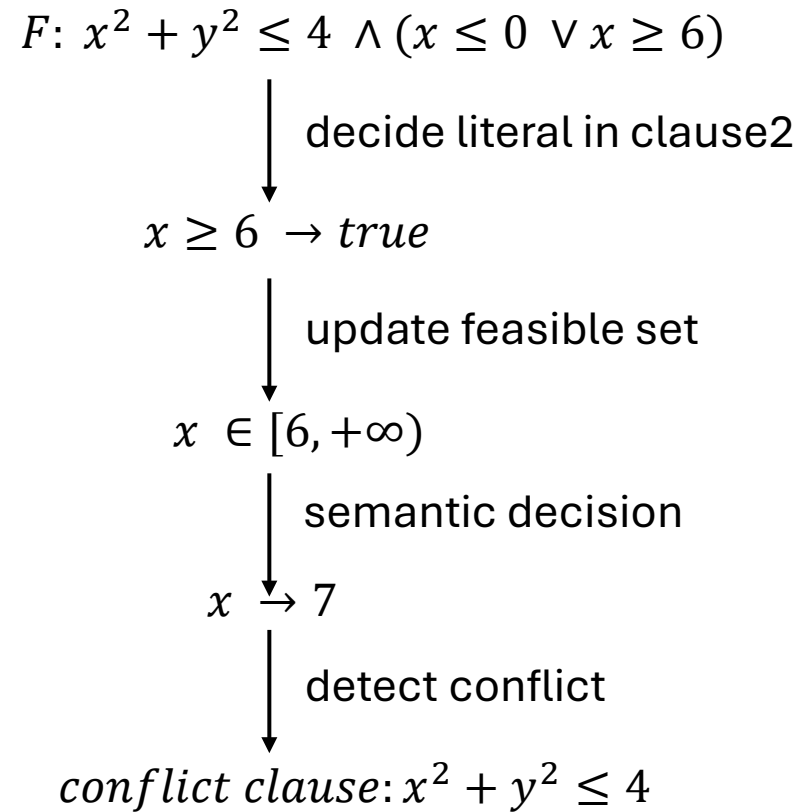
MCSAT/NLSAT Framework

MCSAT searches the solution space by removing conflict spaces
(i.e. learning new lemma)



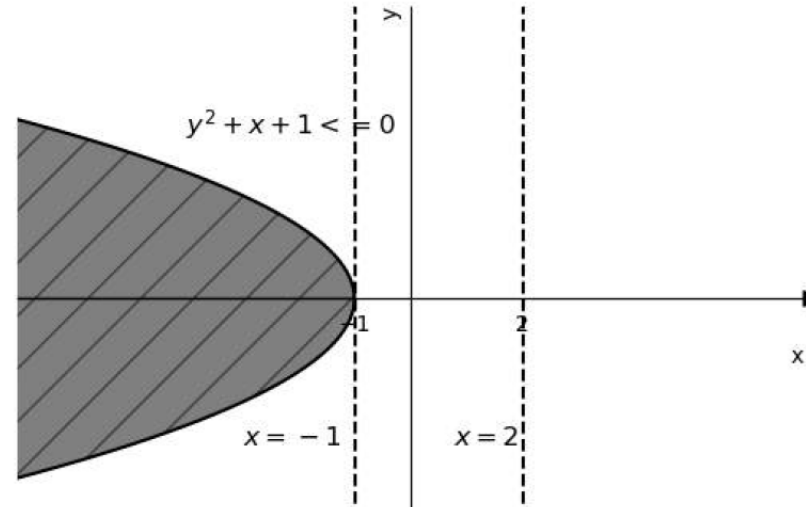
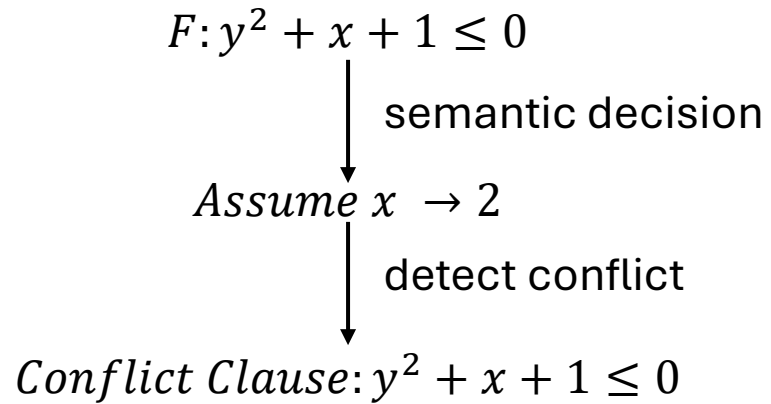
MCSAT/NLSAT Framework

However, MCSAT is short-sighted when facing multiple-literal clauses. **Literal decision is made before feasible set update.**



Conflict Type I: Semantic Based

First case happens when selecting a wrong witness



1. Caused by an incorrect semantic decision.
2. Resolved later by a newly learned lemma.
3. Unforeseeable at the time of decision.

Conflict Type II: Literal Decision

Second case happens when deciding a wrong literal

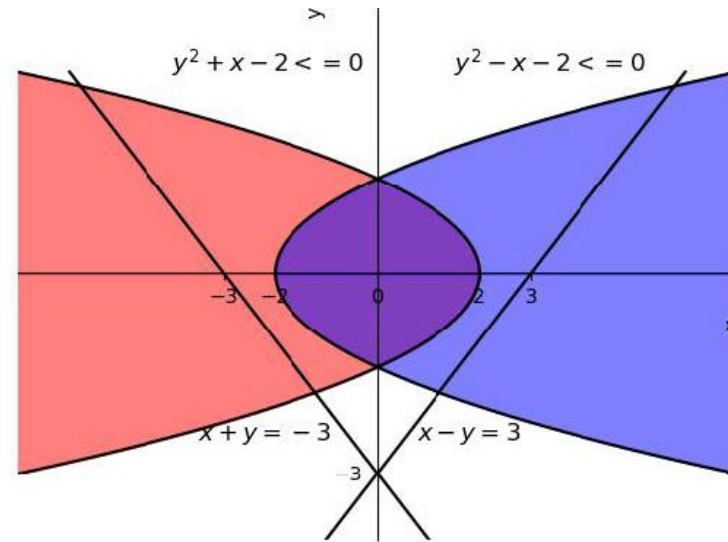
$$C_1: y^2 + x - 2 \leq 0 \vee y^2 - x - 2 \leq 0$$

$$C_2: x + y = -3$$

$$C_3: x - y = 3$$

$$\{C_1, C_2\}: \text{decide } y^2 + x - 2 \leq 0$$

$$\{C_1, C_3\}: \text{decide } y^2 - x - 2 \leq 0$$



Q: How do we dynamically detect literals?

1. Caused by an incorrect literal decision.
2. Could be avoided by proper literal decision strategy.
3. Foreseeable at the time of decision using interval view.

Contribution I: Look-Ahead

Step 1: Compute Clause-Level Feasible Set

Example 5.

$$\left. \begin{array}{l} c_1 : [-4, -2] \vee [2, 4] \rightarrow \{[-4, -2] \cup [2, 4]\}, \\ c_2 : [-6, -5] \vee [1, 5] \rightarrow \{[-6, -5] \cup [1, 5]\} \end{array} \right\} \wedge \rightarrow \{[2, 4]\}$$

This example shows a path case, since the clauses can be satisfied by deciding literals in the green boxes.

Example 6.

$$\left. \begin{array}{l} c_1 : [-4, -2] \vee [2, 4] \rightarrow \{[-4, -2] \cup [2, 4]\}, \\ c_2 : [-6, -5] \vee [5, 6] \rightarrow \{[-6, -5] \cup [5, 6]\} \end{array} \right\} \wedge \rightarrow \emptyset$$

This example shows a block case: the clauses cannot be satisfied regardless of which literals we decide.

Step 2: Select witness within the interval set, for example $x \rightarrow 3$

Step 3: Decide literals satisfied under the chosen witness (green boxes)

Contribution I: Look-Ahead

Algorithm 3: Deciding Literals Using Pre-Appointed Value

Input : A set of clauses F , pre-appointed value val selected from feasible-set

Output: Decided literals $lits$

```
1  $lits \leftarrow \emptyset$ ;
2 for each clause  $c \in F$  do
3    $path\_literal \leftarrow$  undefined;
4   for each literal  $l \in c$  do
5      $lit\_set \leftarrow$  compute feasible-set of  $l$ ;
6      $val\_lit \leftarrow$  real propagate literal  $l$ ;
7     if  $val\_lit = \top$  then
8       break // clause already
        satisfied
9     if  $val\_lit = \perp$  then
10      continue // literal unsatisfied,
        check next literal
11     if  $lit\_set$  contains  $val$  then
12       $path\_literal \leftarrow l$  // decide
        satisfiable literal under
        pre-appointed value
13   if only one literal undefined in  $c$  then
14     unit propagate the literal;
15   else if  $path\_literal$  is defined then
16      $lits \leftarrow lits \cup \{path\_literal\}$ ;
17 return  $lits$ ;
```

Algorithm 4: Process Clauses After a New Lemma

Input: A new lemma $lemma$, arithmetic variable v

```
1  $lemma\_feasible\_set \leftarrow$  compute feasible-set of
   clause( $lemma$ );
2  $feasible\_set[v] \leftarrow$ 
    $feasible\_set[v] \cap lemma\_feasible\_set$ ;
3 if  $feasible\_set[v]$  is empty then
4   // Block case: no consistent
    assignment possible for  $v$ 
    // Proceed as in original NLSAT
    (Algorithm 1)
    original process clauses;
5 else
6   // Path case: a consistent
    assignment exists for  $v$ 
     $val \leftarrow value\_selection(feasible\_set[v])$ ;
    // Call Algorithm 3 to decide
    literals using the pre-appointed
    value
7   deciding literals using pre-appointed value ( $val$ );
```

Propagation in NLSAT

Type 1: Unit Propagation

When literal $x \leq 0$ is decided to be false

$$x^2 + y^2 \leq 4 \vee x \leq 0 \rightarrow x^2 + y^2 \leq 4$$

Type 2: Real Propagation

For real arithmetic

$$x^2 < 0 \rightarrow \textit{false}$$

Both propagation methods remain confined to the literal level, missing the guidance of arithmetic semantics.

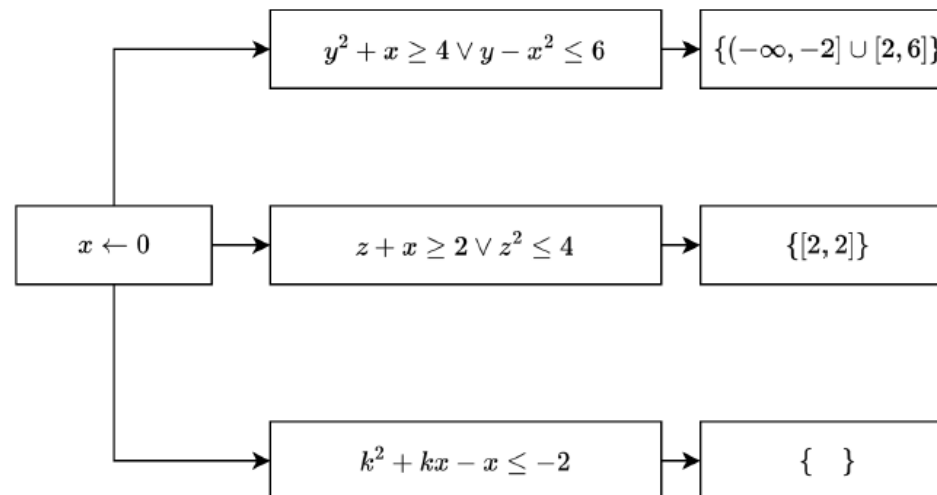
Contribution II :Clause-Level Propagation

Propagation in SAT Solving:

1. Detect conflict or deduce assignment
2. Happens after new assignment

Clause-Level Propagation

Definition 4. *Given a clause c , an arithmetic variable x , and an assignment α that assigns all variables appearing in c except x , clause-level propagation on x is the computation of the feasible-set of the clause c , which serves to narrow the feasible-set of the variable x .*



New Branching Heuristic

Algorithm 6: Propagation-Based Branching Heuristic

Output: A variable v to branch on

```
1 // Prioritize blocked variables
   (potential conflicts) first
2 if  $blocked\_vars \neq \emptyset$  then
3    $v \leftarrow select\_from(blocked\_vars);$ 
4 // Next, consider fixed variables
   (propagate value)
5 else if  $fixed\_vars \neq \emptyset$  then
6    $v \leftarrow select\_from(fixed\_vars);$ 
   // Otherwise, use VSIDS heuristic for
   normal variables
7 else
8    $v \leftarrow vsids\_select();$ 
9 return  $v$ 
```

Evaluation

- SMTLIB QF_NRA benchmark

Category	#inst	Z3	YICES2	CVC5	NLSAT	Ours
20161105-Sturm-MBO	405	SAT	0	0	0	0
		UNSAT	124	285	285	39
		SOLVED	124	285	285	39
20161105-Sturm-MGC	9	SAT	2	0	0	2
		UNSAT	7	0	0	6
		SOLVED	9	0	0	8
20170501-Heizmann	69	SAT	2	0	1	2
		UNSAT	1	12	9	19
		SOLVED	3	12	10	21
20180501-Economics-Mulligan	135	SAT	93	91	89	92
		UNSAT	39	39	35	41
		SOLVED	132	130	124	134
2019-ezsmst	63	SAT	56	52	50	58
		UNSAT	2	2	2	2
		SOLVED	58	54	52	60
20200911-Pine	245	SAT	234	235	199	235
		UNSAT	6	8	5	5
		SOLVED	240	243	204	239
20211101-Geogebra	112	SAT	110	99	91	110
		UNSAT	0	0	0	0
		SOLVED	110	99	91	110
20220314-Uncu	225	SAT	69	70	62	68
		UNSAT	155	153	148	155
		SOLVED	224	223	210	222
hong	20	SAT	0	0	0	0
		UNSAT	8	20	20	14
		SOLVED	8	20	20	14
hycomp	2752	SAT	307	227	225	244
		UNSAT	2242	2201	2212	2088
		SOLVED	2549	2428	2437	2332
kissing	45	SAT	33	10	17	14
		UNSAT	0	0	0	0
		SOLVED	33	10	17	14
LassoRanker	821	SAT	167	122	305	220
		UNSAT	151	260	470	174
		SOLVED	318	382	775	394
meti-tarski	7006	SAT	4391	4369	4343	4391
		UNSAT	2605	2588	2581	2611
		SOLVED	6996	6957	6924	7002
UltimateAutomizer	61	SAT	35	39	35	45
		UNSAT	11	12	10	13
		SOLVED	46	51	45	58
zankl	166	SAT	70	58	58	62
		UNSAT	28	32	32	27
		SOLVED	98	90	90	89
Total	12134	SAT	5569	5372	5475	5541
		UNSAT	5379	5612	5809	5191
		SOLVED	10948	10984	11284	11005

Conclusion

1. New literal decision method: look-ahead
2. New propagation method: clause-level propagation
3. Propagation based branching heuristic
4. SOTA solver for satisfiable instances